

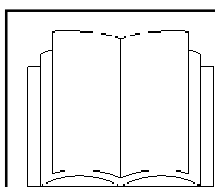
AVANT[®]

Operator's Manual for Attachment



Snow Plow

Product number: 1800 mm plow for 400-700-series loaders: A21048
 2200 mm plow for 400-700-series loaders: A34368



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

Sales:

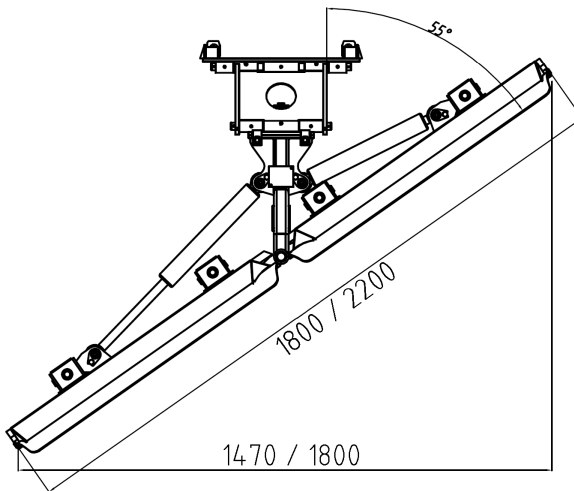
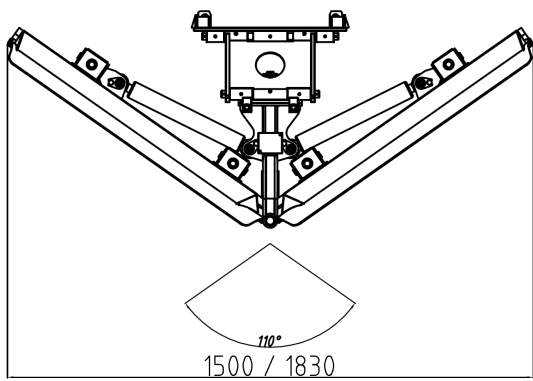
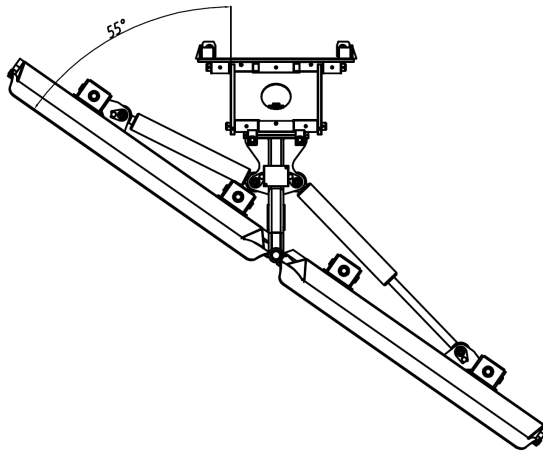
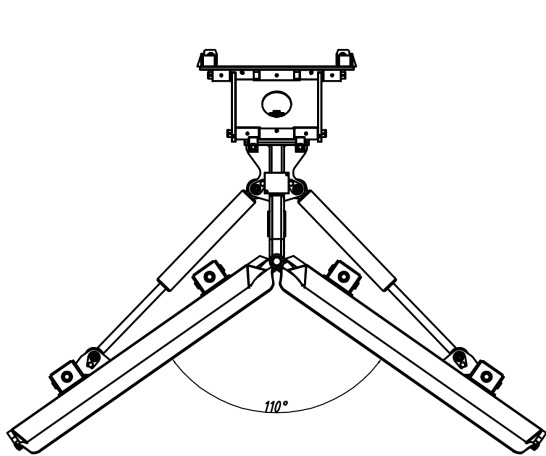
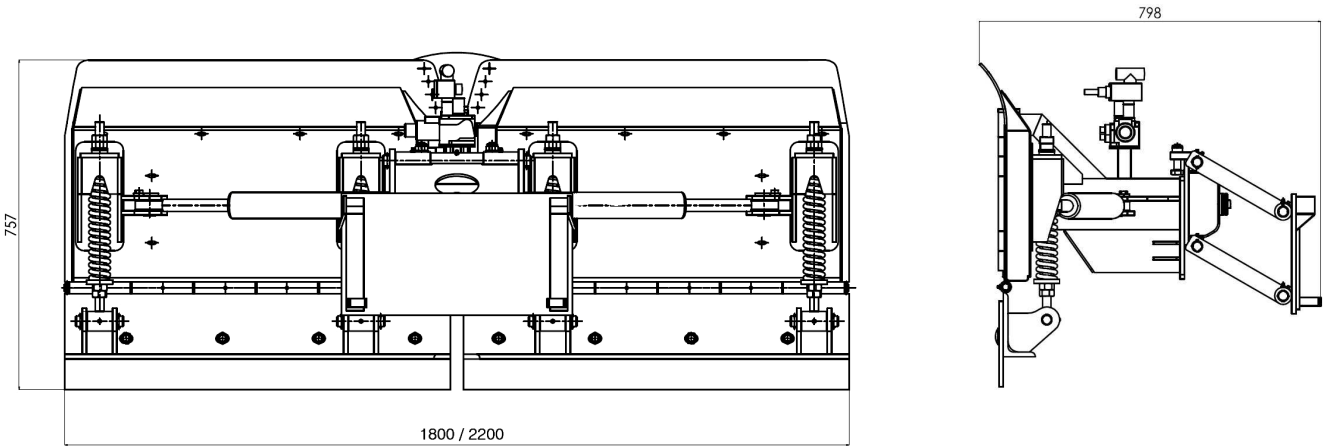


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Appendix: EC Declaration of Conformity

1. Foreword

Avant Tecno Oy and Sami AS would like to thank you for your purchase of this attachment for your Avant loader. It has been designed and manufactured on the basis of years of experience on product development and manufacturing. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The purpose of this manual is to help you to:

- operate the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life

The following warning symbols are used throughout this manual to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	WARNING:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.

NOTICE

This notice mark indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader, each attachment that are used, and all safety instructions before using the equipment. Using the equipment for other purposes or use in any other way than described in this manual is prohibited. Keep this manual at hand throughout the service life of the equipment. If you sell or transfer the equipment, be sure to hand over this manual to the new owner. If the manual is lost or damaged, you can request a new one from your Avant dealer or manufacturer. Due to continuous product development some of the details shown in this manual may differ from your equipment. This manual contains the original instructions in English.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws, and other regulations concerning the use of the equipment. Particularly the regulations concerning the use of the equipment on public road areas must be observed. We reserve the right to change the contents of the manual without notification.

2. Designed purpose of use

The AVANT Snow plow is an attachment suitable for use with AVANT multi purpose loaders that are shown in table 1. The plow is a versatile articulated plow to transferring and clearing snow. The position of the blades can be adjusted independently directly from the driver's seat with the use of electrically controlled hydraulic valve on the plow. This makes it possible to use plow in various configurations, such as a slewable straight dozer blade, collecting v-blade or as a point plow.

The floating mounting bracket allows vertical movement of the plow as well as horizontal twisting for the best possible ground surface following. In addition, the lower blade sections are hinged in a way that allows them to tilt backwards if they hit an obstacle, preventing sudden stopping of the loader and protecting both the driver and the blades. Optional adjustable skids help the plow to move smoothly even on rough surfaces. The snow plow is designed to be installed on an Avant loader for clearing and pushing snow on street areas. Operating any controls of the plow or the loader is permitted only for persons who are familiar with specific safety instructions and are capable to operate the equipment safely. Operating any control is allowed only when sitting on the driver's seat. Using the plow in other way or for other purposes than shown in this manual is prohibited.

The attachments have been designed to require as little maintenance as possible. The operator can perform regular maintenance tasks. All repair work can't normally be performed by the operator, and demanding repair and maintenance tasks, such as servicing or replacing hydraulic components, should left for professional service technicians. All maintenance work must be done using proper safety equipment. All spare parts must be similar to original specifications, which can be ensured by using only original Avant spare parts.

Familiarise yourself with the instructions regarding service and maintenance in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment, or if you require spare parts or maintenance services.

Table 1 – Snow Plows A21048 and A34368 compatibility with Avant loader models

Model	216 218 220	313S 320S 320S+	419 420	520 528 525LPG	R28 R35	630 635 640	745 750
A21048	-	-	•	•	•	•	•
A35368	-	-	(•)	•	•	•	•

When using with models marked with (•), full performance may not be achieved. Additional counterweights may be required. For compatibility with a model not shown in table, contact your Avant retailer.

3. Safety instructions for using the Snow Plow

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



Misuse, careless use, or using an attachment that is in poor condition, may cause serious injuries. Familiarise yourself with the controls of the loader and coupling of the attachment at a safe area. Study especially how to stop the equipment in a safe manner. Read all safety precautions carefully.

Read all safety instructions carefully before handling the attachment



- When attaching the attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are positively in the lower position and that they have locked the attachment to the loader.**
- The plow is intended to be used by one operator at a time. Do not allow others to approach the area where the machine is in operation.
- Do not drive with the plow at high speed, unless you are sure, that there are no obstacles in the operating area. The loader may stop suddenly if the plow hits an obstacle, regardless of the hinged blade sections. The hinged plow sections do not replace safe and correct operating practices. Keep in mind that obstacles or fragile surfaces may be hidden under snow. Keep the hinged plow sections in good condition.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Pay attention to the contours of the terrain as well as other hazards, such as loose rocks, or branches and trees that reach the driver's area.
- Always transport the plow as low and as close to the machine as possible to keep the centre of gravity low. Operate the controls of the loader slowly and in calm manner especially during plowing.
- Lower the plow firmly on the ground before leaving the driver's seat. Make sure that the attachment is properly supported during any maintenance or inspections. Do not leave the driver's seat when the loader boom is lifted. Going under a raised attachment or loader boom is dangerous, as the boom may lower due to loss of stability, mechanical fault or when other person operates the controls of the loader.
- Be sure not to tilt the attachment too far up to prevent the plow from tilting over the tipping point of the hinged floating quick attach plate.
- Make sure to use only an attachment that is in good condition. Perform daily inspections and read the instructions regarding service and maintenance. Never operate the attachment if the hydraulic systems of both the attachment and the loader are not completely intact.
- Shut down the loader engine and release any residual pressure before adjustments or maintenance work.
- Do not modify the attachment in a manner that would affect its safety. To make sure that the plow is safe, use only specified spare parts.
- Make sure that the loader is equipped with necessary safety equipment, and that they are in order. Seat belt must be used. If there are specific risks at the operating area, appropriate safety equipment must be used.
- Also read the safety instructions of the loader from the operator's manual of the loader. Get familiar with the controls of the loader and how to drive it safely, and also note the effects that slippery surfaces pose to the handling characteristics of the loader. On slippery areas drive calmly and deliberately.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or attachment.



Familiarize yourself with the controls and handling of the loader at a safe area. Pay particular attention to safe stopping of the loader. Note that the handling characteristics of the loader may be different on slippery surfaces and cold conditions. Allow the loader to warm up before use.

Remember to wear proper protective clothing:



- The noise level at the driver's seat may exceed 85 dB(A) depending on loader model and working conditions. Wear hearing protection while working with the loader.



- Wear safety boots when working with the loader.



- Wear protective gloves. When handling hydraulic hoses or components, wear also safety goggles.

4. Specifications

Table 2 – Snow Plow technical specifications

Product number:	A21048	A34368
Working width:	1500 - 1800 mm	1800 - 2200 mm
Total weight:	280 kg	305 kg
Blade height:	770 mm	
Slewing angle:	±35° each blade	
Safety features:	Plow wing lower sections spring loaded, plow wing hydraulic system allows cylinders to compress	
Optional support legs with skid shoes:	A417068	
Maximum allowed input of hydraulic energy:	22,5 MPa (225 bar) 60 l/min	
Suitable AVANT loader:	See Table 1	

4.1. Safety labels and main components of the Snow Plow

Listed below are the labels and markings, which must be visible on the equipment. Replace any warning label that has become unclear or has detached completely. New labels are available via your retailer or contact information provided on the cover of this manual.



The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace damaged or missing warning labels with new ones.



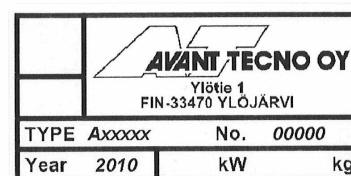
A46771



A46772



A46772



Identification plates of the attachment

Table 3 – Decal warning messages and locations

Decal	Explanation
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or loader boom; stay away from raised equipment.
A46803	Crushing hazard, keep clear of moving components, floating mounting bracket and turning blade sections.
	Attachment identification plates

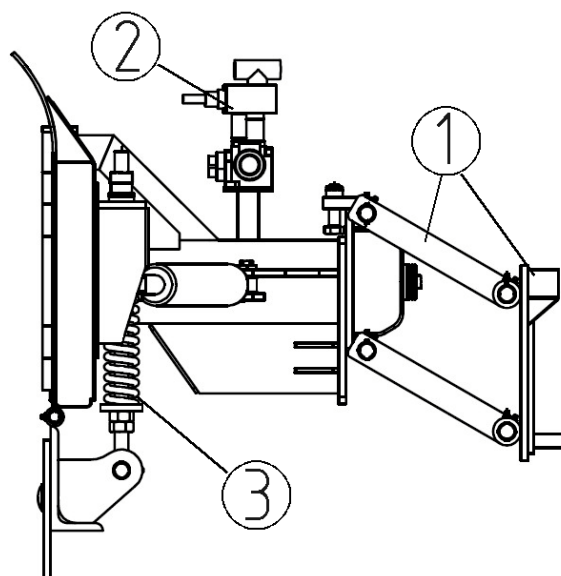
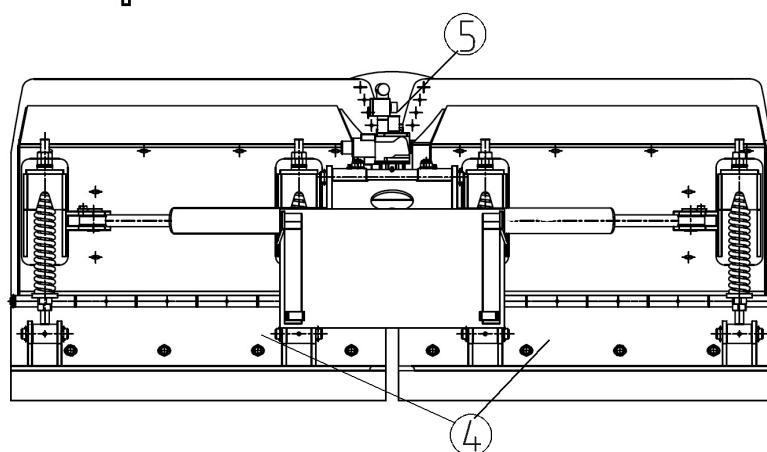


Table 4 – Main components of the snow plow

1	Quick coupling plate and floating mounting
2	Electric control valve
3	Lower blade section springs.
4	Hinged lower blade sections
5	Blade pivot

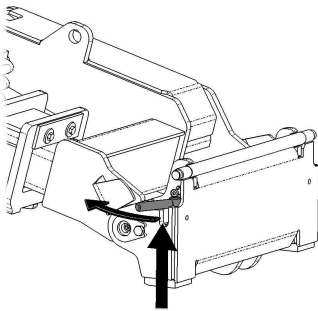


5. Assembly of the Snow Plow

The snow plow is ready for use as delivered. Mount the attachment and its hydraulic hoses to the loader as instructed. Read the instructions related to using the equipment and begin the practise of use at a safe area.

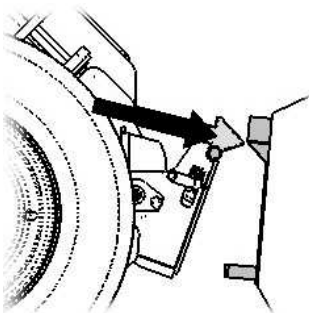
5.1. Mounting the snow plow to the loader

Mounting the attachment to the loader is quick and easy, but it must be done carefully. An attachment that has not been locked may come off from the loader and cause hazardous situations. The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3 and the instructions provided in the operator's manual of the loader.



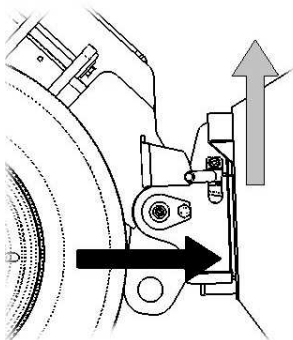
Step 1:

- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses or the electric cables are not in the way during mounting.



Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this. Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.

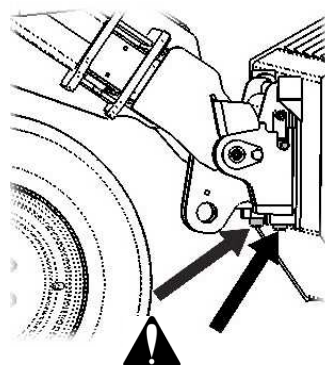


Step 3:

- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually or lock the hydraulic locking.
- Always ensure positive locking of pins.



An attachment that has not been completely locked to the loader may fall on the boom or operator, or under the loader during driving, causing loss of control of the loader. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or lift an attachment that has not been fully locked.



Never use the automatic locking of the locking pins if the plow is lifted to a height of over one metre. The floating mechanism of the attachment may allow the plow to tip over when tilted too much. It is recommended to always lower the attachment on the ground and secure the locking manually.

5.2. Connecting and disconnecting hydraulic fittings

On 400 to 700 series loaders, the multiconnector type hydraulic coupling is used to connect hydraulic hoses. If you want to change the type of the fittings on the loader or on the attachment, contact your Avant dealer or service representative for instructions or installation service. At first, the hydraulic hoses or cylinders may have air trapped in them, causing inaccurate movements of the plow. The system is self-bleeding when operating the plow to its extreme positions; separate procedure is not needed to remove air.

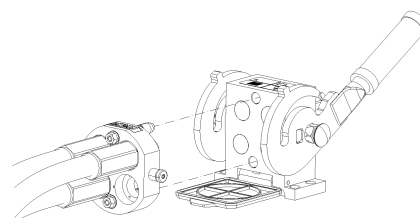


Never connect or disconnect quick couplings or other hydraulic components if the system is under pressure, or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious burns or other injuries.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



To disconnect the multiconnector system, turn off the auxiliary hydraulics of the loader, and unlock the multiconnector. To unlock, push the knob on the side of the connector, and turn the lever to disconnect.

Conventional quick couplings (alternative couplings for some loader models):

Before connecting or disconnecting the standard quick couplings, the residual pressure must be released as shown below. The conventional quick couplings will not connect, if there is pressure in the hydraulic system.

To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



NOTICE

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

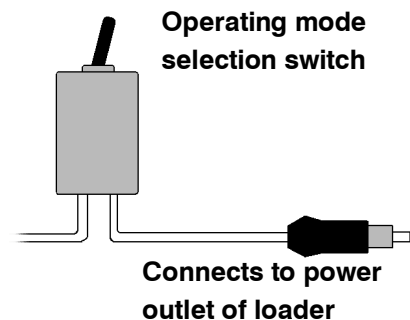
Releasing residual hydraulic pressure:

To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader to it's extreme positions before disconnecting the couplings. If the fittings will not connect to the loader, the residual pressure must be released. The residual pressure must be released before handling or going near the moving components of the attachment and before storage of the attachment.

5.3. Electric connection

To select the operating mode of the plow, there is a hydraulic valve on the plow. The electric harness of the valve is connected to the loader. There are two alternatives for connecting the electric harness:

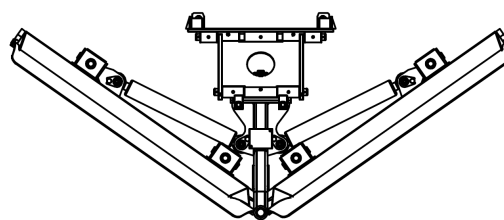
- The optional attachment control switch pack, available for 400-700 loader models, is recommended. In this case, the connector of the attachment is connected to the socket located next to hydraulic connectors on the loader boom. The separate cable set is not needed when using attachment control switch pack. To control the blade, the switch number 1 is used, other switches are not in use.
- Alternatively the cable harness A35358, delivered with the attachment and equipped with a two-position switch, is connected to the power plug near the control panel of the loader. Select the location of the switch in the cab to ensure that it is easily accessible but will not be used inadvertently.
- The operating modes and controls are shown in chapter 6.2.2.



Route the electric cable so that it cannot get pinched, stuck, or stretched. Place the switch in the cab so that it is easily accessible but will not be used inadvertently. With the help of the attachment control switch pack available as an option for the loaders, the attachment control switches are easily accessible at the end of the boom control lever. Instructions covering the loader plug are provided in the operator's manual of the loader.

5.4. Uncoupling and storage of the snow plow

Before unmounting the plow set it to the position shown in adjacent figure. Release and store the plow on a level surface so that the plow will stand firmly on both of its blade sections. Avoid leaving the plow in direct contact against the ground, as the plow may get frozen stuck, or rust damage may develop. Place a pallet or blocks of wood under the plow, for example. The best storage place for the attachment is an area that is protected from rain and direct sunlight. Remove most of the snow from the top of the plow to avoid the plow from becoming stuck by re-freezing snow on the blades, melted by the warm hydraulic oil of the loader. Ensure the stability of the plow during its storage. Climbing on the snow plow is prohibited. Service the attachment before longer storage periods, read instructions concerning maintenance and service from Chapter 7.



Storage position of the snow plow

NOTICE

When uncoupling the attachment from the loader, always disconnect the hydraulic and electric couplings first before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent dirt from entering the hydraulic system.

6. Instructions for use

After the snow plow has been mounted as described in previous chapter, it is ready for use. Check the attachment and the operating environment once more before starting to work. Quick inspection of the snow plow and the operating area prior use are part of ensuring safety and the best performance of the equipment.

6.1. Checks before use

- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. The attachment must not be used, if there is a fault in the hydraulic system of the loader or the attachment. Refer to Chapter 7 for maintenance instructions.
- Check the condition of the plow blades and their lower sections. Check that the lower sections of the blades can move and that the spring tension system
- Check the condition of the mixing and spreading shafts, and that they can rotate freely. Check that there is no string or other foreign material coiled around them.
- Check that the drive chain cover is securely in its place on the side of the spreader. Using the spreader without the guard in its place is prohibited. Check the drive chain if the spreader makes loud noise during operation.
- Lift and lock the outriggers to upper position always when using the spreader.
- Before starting work, remove larger obstacles from the working area. Remember correct working methods and avoid leaving the driver's seat unnecessarily.
- Drive carefully on uneven terrain and note the effect of heavy load at the rear or at the front of the loader to the stability and handling of the loader.
- Ensure that the loader and the attachment are used in a safe and intended manner. Operate the controls of the loader only when sitting on the driver's seat. Do not allow children to operate the machine.



Check the operating area before using the equipment:

- Make sure, that there are no hidden obstacles, for example, under the snow. Running into an obstacle may stop the machine suddenly and unexpectedly regardless of the safety features of the plow. Drive slowly to avoid hazards related to sudden stopping of the loader. Always wear the safety belt when operating the loader.
- Ensure that there is nobody else in the working area. Do not allow assisting persons to go directly in front of the snow plow or to reach the boom danger zone.
- Also ensure that there is nobody in the way when reversing with the loader. The safety features of the plow function only when driving forward.

6.2. Operating the snow plow

The versatile plow and separately controllable blade sections make it possible to use the plow efficiently at different operating areas and in a way that best suits the operating conditions

- Using the plow as a point plow makes clearing narrow roads or sidewalks with single drive, and makes it possible to plow a path even on thick snow.
- The tilted position can be adjusted as needed according to snow layer thickness and needed plowing width. The side where the plow is tilted can be changed very fast.
- Operating the plow as a collecting plow makes it easier to stack snow, and works as an excellent tool to help finalizing plowing.

The correct operating position of the plow has a great effect on the operation of the plow. Familiarize yourself carefully with the controls and correct position of the plow, to be able to use the plow in the most efficient way.

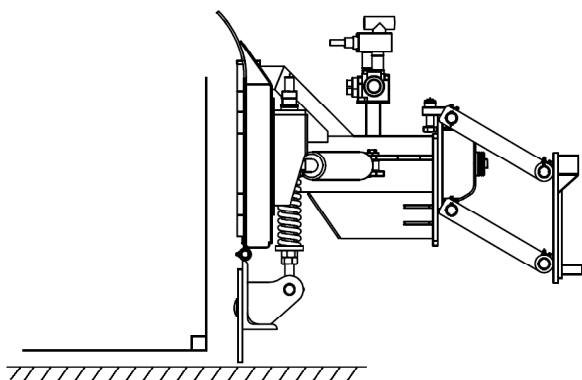


Make sure not to lift or tilt the plow too high. Tilting the plow too much when lifted high may cause it to tip over on the loader boom, as the floating attachment mechanism on the plow allows it to move. Always lower the plow on the ground before leaving the driver's seat.

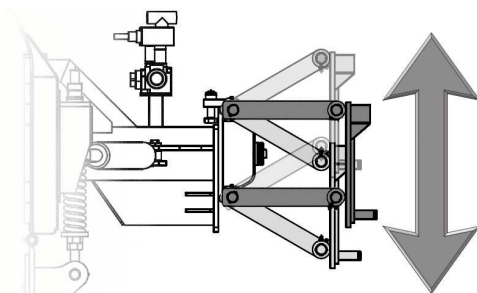
6.2.1. Correct operating position

For the best performance of the plow, its position must be set so that the plow is in upright position.

1. Set the telescopic boom of the loader to the position in which you intend to operate the plow
2. Set and tilt the blade on the ground into upright position by using the boom control lever of the loader. Set the height of the boom so that the floating attachment bracket will be set to its middle position as shown in figures below. If necessary, tilt the attachment again to set it upright.
3. If the loader is equipped with the optional boom floating system, it should be switched on after the plow is set to operating position. The floating bracket of the attachment makes it possible to use the plow efficiently even if the loader is not equipped with boom floating system.



Keep the plow in upright operating position



Set the height of the loader boom so that the bracket is in the middle of its extreme positions

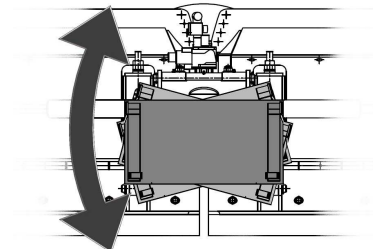
NOTICE

The correct operating position is essential for efficient plowing and controllability of the loader. Keep the plow lightly against the ground in upright position. If the plow is pressed down too much, the steerability of the loader may weaken, as the front tires tend to lift off the ground.

NOTICE

After the correct operating position is set, the best way to lift the plow is to use only the lifting and lowering movements of the loader boom. In this way the plow will stay upright and ready to use when the plow is lowered by simply pushing the loader boom control lever forward.

The floating attachment bracket will allow the plow to also swivel sideways. This makes it possible to lower the plow quickly to its efficient upright position also when operating on uneven terrain.

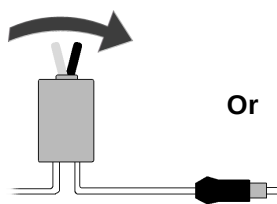


Plow swivels also sideways

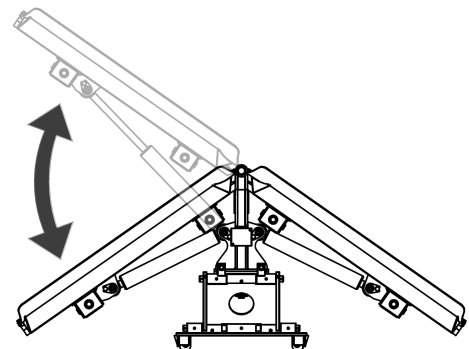
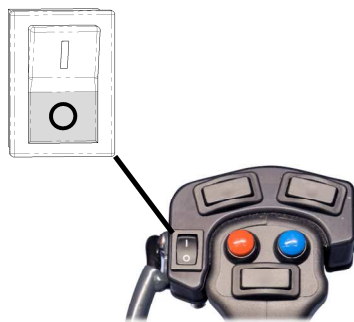
6.2.2. Turning plow blade sections

The desired operating mode is selected with either of the electric coupling alternatives, and after that, the plow is operated with the auxiliary hydraulics control lever or the electric joystick buttons of the loader. See Chapter 5.3. for instructions about the electric connection. If the electric harness is not connected, only the left blade section can be controlled as shown below.

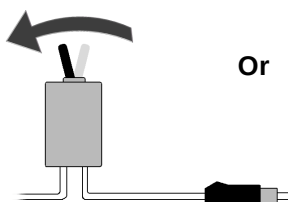
- When the electric switch is in 0-position, the auxiliary hydraulics control lever of the loader controls the left blade section of the plow independently.



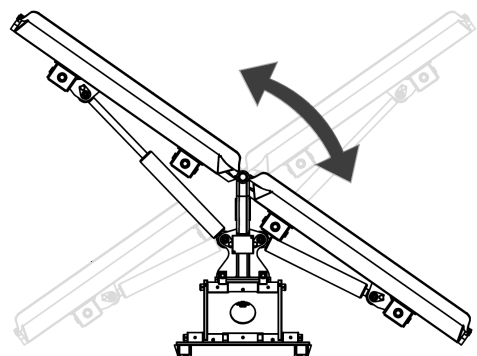
Or



- When the electric switch is in 1-position, the auxiliary hydraulics control lever of the loader controls the whole plow. If either of the blade sections is turned to its extreme position, the plow will continue to turn, until the other section also reaches its extreme position.



Or



Safety features of the plow:

To reduce the possibility of a sudden stop after hitting an obstacle, the plow is equipped with hydraulically yielding blade sections and spring tensioned lower sections of the blade, that allow the blades to move backwards. The hydraulic system of the low includes a pressure relief valve which will allow the blade sections to yield in hard collisions with obstacles, is the blade sections are not in their extreme rear positions. Changing the adjustment of the system is prohibited, contact authorized service point if encountering problems with the system. The lower blade sections are held by compression springs. The lower sections will move slightly during normal use, and the movement must not be blocked and the springs must not be tensioned excessively.



The safety features of the plow reduce the risk of sudden stopping of the machine and the possibility of injuries that could follow. However, the existence of the safety features will not replace the careful and cautious operating, and the safety features are not enough to guarantee safe operation of the plow. The performance of the safety features may be hindered in certain conditions. Always use seat belt and drive slowly, and remove or mark any obstacle in the operating area so that collisions can be avoided.



The plow is intended to be used only when driving forward, as the safety features are operational only in forward travel. When reversing, the plow is fixed, causing sudden stop if the blade hits any obstacle or gets stuck on the ground surface.

7. Maintenance and service

The attachment is designed to be as maintenance free as possible. Continuous maintenance of the snow plow includes regular cleaning, monitoring the condition of the blades and lubricating. If there are problems on the snow plow that can't be solved with the instructions shown in this manual, contact your nearest Avant dealer or authorized service point.



Never go under a raised attachment or the loader boom. The loader boom will move down and may be tilted even when the engine has been shut down. Ensure that the attachment is properly supported during any maintenance procedure. Use the service bracket provided with the loader to support the boom and use also additional means as necessary.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a leak in the hydraulic system of the attachment or the loader. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid penetrates the skin. Hydraulic fluid is also harmful to the environment. Operate only with hydraulic oil that is accepted for use in Avant loaders.



Never handle a hydraulic system that is pressurized, since a loosening of a fitting or component failure may release powerful thin spray of oil, which can penetrate skin and cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. The hoses have limited service life, which is marked on the hoses. After the date marked on the hoses, they should be replaced with new ones.

If there are signs of leaks, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component must be replaced. Spare parts are available from your nearest AVANT retailer or service point. Leave the repair work to professional service technicians, if you don't have experience about hydraulic assemblies and how to perform the repairing safely.

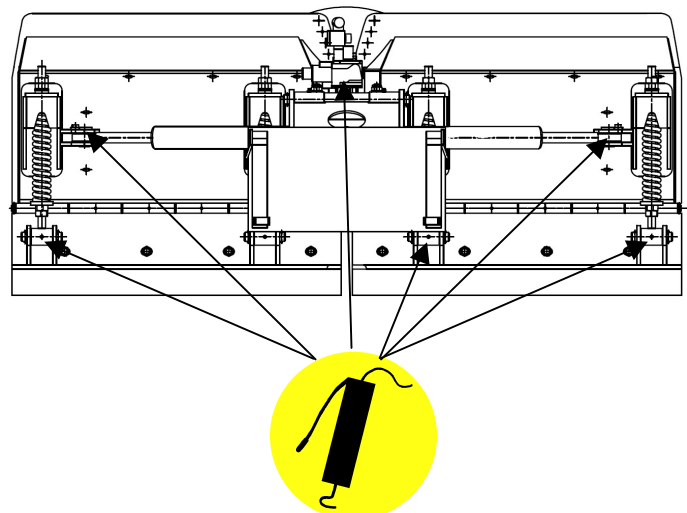
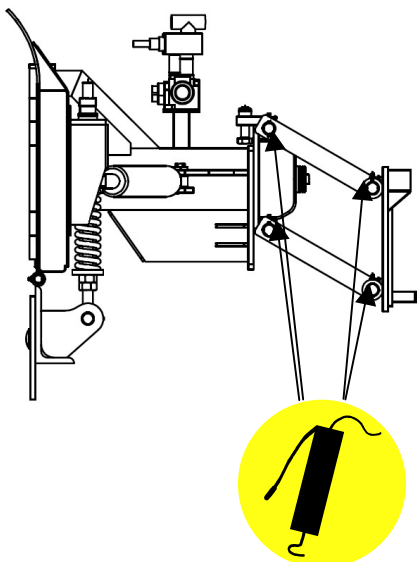
7.1. Cleaning and lubrication

You may use a pressure washer and a mild detergent to wash the equipment. Avoid spraying directly at the hydraulic components or using solvents when washing. In addition, mind the labels on the attachment. Lubricate the lubrication points after washing. There are several lubricating points on the snow plow, located in the hinges of the floating attachment bracket, plow pivot hinge, and at the ends of the hydraulic cylinders. The lubricating points can be lubricated with a standard grease press.

Clean the attachment thoroughly before long storage periods, touch up the paint where needed, and lubricate lubricating points to prevent rust damage. Protect the visible part of the cylinder shafts by applying a small amount of oil to them

NOTICE

Lubricant must be added about 1 to 3 times a year. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

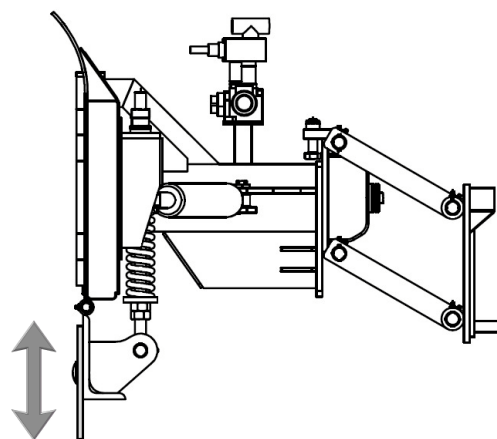


7.2. Adjusting the snow plow

The lower blade sections should be adjusted so that the frame of the plow will not touch the ground.

NOTICE

To prevent wearing or damaging the plow, the frame must not contact the ground continuously. Adjust or replace the lower blade sections before the frame touches the ground.



7.3. Servicing safety features

The lower blade sections will move during normal operation and their movement should not be blocked or restrained. Do not over tighten the compression springs. If there are problems with the blade section tilting, contact the service department. When handling the compressed springs, the hazards related to them must be observed. Checking and adjusting the hydraulic safety relief valve must be left for authorized service point, if there are any problems related to its operation.



The springs of the plow are tensioned and careless maintenance could lead to serious injuries if the spring is released. Do not disassemble the tensioned springs of the snow plow; leave the maintenance for professional service technicians.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY -vaatimustenmukaisuusvakuutus EC Declaration of Conformity

Valmistaja / Manufacturer :

AS SAMI

Osoite / Address :

Tule 20

76501 Saue, HARJUMAA, ESTONIA

www.sami.ee

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
EN ISO 12100, EN ISO 4413, EN 13021

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
EN ISO 12100, EN ISO 4413, EN 13021

Kuvaus:

Avant-kuormainten kanssa käytettävä lumiaura

Description:

Snow Plow; attachment for Avant loaders

Mallit / Models :

Avant		400-700 series
Alueaura	1800 mm	A21048
Snow Plow	2200 mm	A34368

Allekirjoitus / Signature :


Gert Jugala,
Johtaja/
Director

20.06

2011 Saue, Estonia





AVANT[®]